

Title

Immunological correlates and subgroup efficacy analysis of monoclonal antibody L9LS for malaria: findings from the extended follow-up

Abstract

Background: Monoclonal antibody efficacy may vary by host and transmission factors. We conducted secondary analysis of immune responses and treatment heterogeneity. **Methods:** Serum samples and clinical data from 324 trial participants were analyzed. Circulating antibody titers, cellular immune markers, and age/nutritional subgroups were examined in relation to parasitaemia outcomes. **Results:** Antibody persistence at 12 months was higher in the two-dose group (median titer 850 versus 340 in single-dose; $p < 0.001$). In stratified analyses, efficacy was greater in children aged 24-59 months (protective efficacy 49%) than 5-23 months (33%). Baseline nutritional status and prior malaria exposure did not significantly modify efficacy. Clinical malaria episodes (fever with parasitaemia) were reduced 38% in two-dose versus placebo. **Conclusions:** Antibody concentration and age appear to influence L9LS protective efficacy; optimized dosing for younger infants and strategies to enhance immune persistence merit investigation.